

ISSN 1606-9277

# TECHNICAL JOURNAL

Vol. 17 No. 01 JUNE 2025

**RIVER RESEARCH INSTITUTE**  
**MINISTRY OF WATER RESOURCES**  
**GOVT. OF THE PEOPLE'S REPUBLIC OF BANGLADESH**

## EDITORIAL BOARD

### Chief Editor



**S M Abu Horayra**  
Director General  
River Research Institute

### Executive Editor



**Dr. Fatima Rukshana**  
Principal Scientific Officer  
Geotechnical Research Directorate, River Research Institute

### Member



**Dr. Moniruzzaman Khan Eusufzai**  
Principal Scientific Officer  
Dhaka Office, River Research Institute



**Engr. Md. Johurul Islam**  
Principal Scientific Officer  
Hydraulic Research Directorate, River Research Institute



**Engr. Md. Zubayerul Islam**  
Senior Scientific Officer  
Geotechnical Research Directorate, River Research Institute



**Physicist Nayan Chandra Ghosh**  
Senior Scientific Officer  
Dhaka Office, River Research Institute



**Md. Moniruzzaman**  
Senior Scientific Officer  
Geotechnical Research Directorate, River Research Institute



**Engr. Omar Al Maimun**  
Senior Scientific Officer  
Hydraulic Research Directorate, River Research Institute



**Engr. Bikash Roy**  
Senior Scientific Officer  
Hydraulic Research Directorate, River Research Institute



**Mst. Shamima Akter**  
Librarian  
Administration and Finance Directorate, River Research Institute



**Engr. Md. Masuduzzaman**  
Assistant Programmer  
Administration and Finance Directorate, River Research Institute

## Reviewers Panel of Technical Journal - River Research Institute



**Dr. Md. Munsur Rahman**

Professor

Institute of Water and Flood Management

Bangladesh University of Engineering and Technology, Dhaka



**Dr. Mehedi Ahmed Ansary**

Professor

Department of Civil Engineering

Bangladesh University of Engineering and Technology, Dhaka



**Dr. Md. Shahidul Islam**

Professor and Chairman

Department of Geography & Environment

University of Dhaka, Dhaka



**Dr. Nasreen Jahan**

Professor and Head

Department of Water Resources Engineering

Bangladesh University of Engineering and Technology, Dhaka



**Dr. Sajal Kumar Adhikary**

Professor

Department of Civil Engineering

Khulna University of Engineering & Technology, Khulna



**Dr. Aysha Akter**

Professor and Head

Department of Civil Engineering

Chittagong University of Engineering & Technology, Chittagong



**Dr. Mohammed Mizanur Rahman**

Professor

Department of Irrigation and Water Management

Bangladesh Agricultural University, Mymensingh



**Dr. Md. Rokonzaman**

Professor

Department of Civil Engineering

Khulna University of Engineering & Technology, Khulna



**Dr. G. M. Sadiqul Islam**

Professor

Department of Civil Engineering

Chittagong University of Engineering & Technology, Chittagong



**Dr. Mohammad Amir Hossain Bhuiyan**

Professor

Department of Environmental Sciences

Jahangirnagar University, Dhaka

**TECHNICAL JOURNAL**  
**River Research Institute**  
**Vol. 17, No. 01, June 2025**

**Table of Contents**

| <b>Sl. No.</b> | <b>Title of the Paper with Author's Name</b>                                                                                                                                                                                                                                                      | <b>Page No.</b> |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1              | <b>Estimating Suspended Sediment Concentration using Acoustic Doppler Current Profiler (ADCP)</b><br><i>Sumiya Ferdhous, Bikash Roy and Md. Moniruzzaman</i>                                                                                                                                      | 1-7             |
| 2              | <b>Irrigation Challenges and Opportunities in the selected Polders of Coastal Bangladesh</b><br><i>Md. Mostafizur Rahman</i>                                                                                                                                                                      | 8-19            |
| 3              | <b>Modeling Fish Habitat Suitability in the Karnafuli River, Bangladesh</b><br><i>Aysha Akter and Ahammed Dayem</i>                                                                                                                                                                               | 20-31           |
| 4              | <b>Environmental Flow Estimation for the Jamuna river</b><br><i>Sheikh Azmaeen Jawad, Mairukh Wardi and Abul Fazal M. Saleh</i>                                                                                                                                                                   | 32-42           |
| 5              | <b>Performance Analysis of Water Intake System in the Ganges River at Godagari using Physical Model</b><br><i>Bikash Roy, Omar Al Maimun, Md. Johurul Islam, Md. Shahabuddin, Md. Moniruzzaman, M. Masuduzzaman and Pintu Kanungoe</i>                                                            | 43-54           |
| 6              | <b>Channel Planform Dynamics of Bangladesh's Major Rivers: a Study on Braiding and Sinuosity Indices</b><br><i>Md. Moniruzzaman, Md. Masuduzzaman, Sumiya Ferdhous and Fatima Rukshana</i>                                                                                                        | 55-65           |
| 7              | <b>A Hydro-Morphological Study of the Proposed Bridge on the Pandab-Paira River under Patuakhali Road Division</b><br><i>A K M Ashrafuzzaman, Md. Johurul Islam, Md. Tofiquzzaman, Fatima Rukshana, Md. Shahabuddin and Pintu Kanungoe</i>                                                        | 66-76           |
| 8              | <b>Investigating the Trends of Rainfall and Discharge along the Brahmaputra-Jamuna River System</b><br><i>Raied Ahmed Nishat and Mahfujur Rahman Joy</i>                                                                                                                                          | 77-85           |
| 9              | <b>Dynamic Shifting and Future Offtake Formation of the Old Brahmaputra River: A GIS and Remote Sensing Approach</b><br><i>Md. Emran Ali Mondal, Md. Azizul Haque Podder, Sajia Afrin and Abdullah Al Imran</i>                                                                                   | 86-97           |
| 10             | <b>Linking River Morphodynamics and Land Use Land Cover (LULC) Changes: An Assessment of Riverbank Dynamics of the Bishkhali River</b><br><i>Md. Fuad Hasan, Marzia Sultana Shefa, Md. Jahid Hasan, Rashidul Islam, Khandker Mostafa Kamal, Fatima Rukshana, Ayesha Akter and Md. Shamsuzzoha</i> | 98-108          |
| 11             | <b>Evaluation of Management Measures to Develop a Drainage Strategy for Kallayanpur Catchment, Dhaka, Bangladesh</b><br><i>Md. Mahedi Bin Mostafa and Md. Sabbir Mostafa Khan</i>                                                                                                                 | 109-118         |

| <b>Sl.<br/>No.</b> | <b>Title of the Paper with Author's Name</b>                                                                                                                                  | <b>Page<br/>No.</b> |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| *                  | <b>Physical Modelling for the Improvement of Barisal Harbour Area</b><br><i>Md. Rafiqul Alam, Md. Toufiquzzaman, Khondokar Rajib Ahmed, Sajia Afrin, and Md. Moniruzzaman</i> | **<br>11-25         |

**Note:** The last article is a replacement of authorship, published in *Tech.J.RiverRes.Inst.*: 13(1): 11-25, October 2016, without any changes to the original content (\* and \*\*).

